

## HOW BIG IS THE CEREBELLUM?

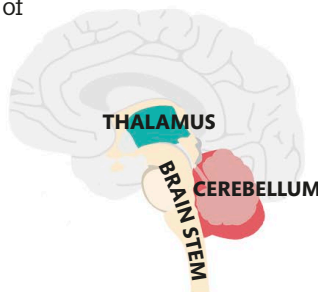
Most of the brain's cells are located in the cerebellum, although it makes up only around 10 percent of the volume of the whole brain.

### Connecting the brain

The stalklike brain stem forms a link between the thalamus, the base of the forebrain, and the spinal cord, which connects to the rest of the body. It is involved in many basic functions, including the sleep-wake cycle, eating, and regulating heart rate.

### The brain stem

The brain stem is made up of three components, all of which have an essential role in several of the human body's most fundamental functions. The midbrain is the start point of the reticular formation, a series of brain nuclei (see pp.32–33) that run through the brain stem and are linked to arousal and alertness and play a crucial role in consciousness. The pons is another series of nuclei that send and receive signals from the cranial nerves associated with the face, ears, and eyes. The medulla descends and narrows to merge with the uppermost end of the spinal cord. It handles many of the autonomous body functions, such as blood-pressure regulation, blushing, and vomiting.



Thalamus links brain stem with forebrain, relaying and preprocessing sensory and other information

THALAMUS

Midbrain is associated with control of state of arousal and body temperature

MIDBRAIN

Pons is a major communication pathway that carries cranial nerves used for breathing, hearing, and eye movements

PONS

BRAIN STEM

10 pairs of cranial nerves emerge from brain stem

Cranial nerves start and end at nuclei in brain stem

MEDULLA

Medulla is involved in important reflexes such as breathing rate and swallowing

Spinal cord consists of a bundle of nerve axons that connect to peripheral nervous system

SPINAL CORD

# The Brain Stem and Cerebellum

The lower regions of the brain are the brain stem, which connects directly to the spinal cord, and the cerebellum, located directly behind it.